Empirical Analysis of Learning Dynamics:
Double Descent, Neural Collapse and CKA

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Presentation Agenda

1. Theoretical Introduction
2. Experimental Setup
3. Results Analysis (Exp 1, 2, 3)
4. Representations Analysis (CKA)
5. Conclusions

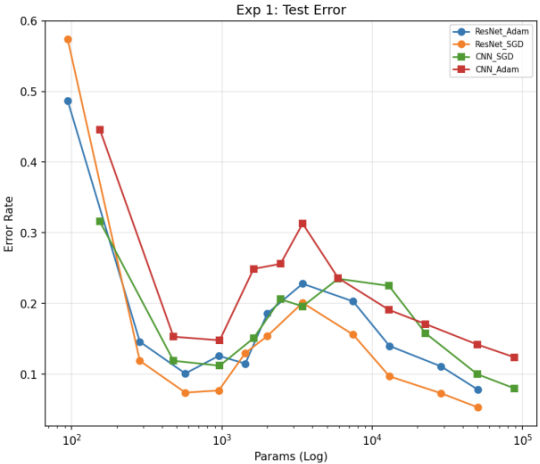
The Modern Paradigm: Deep Double Descent

Classical Theory:

- "U"-shaped Bias-Variance Trade-off.
- Complex models memorize noise ("overfitting").

The Reality of Deep Learning:

- Past the **interpolation threshold**, the test error starts decreasing again.
- Massively over-parameterized models generalize excellently.



What is Neural Collapse (NC)?

In the terminal phases of training, last-layer representations exhibit 4 geometric properties (Papyan et al., 2020):

1. **NC1 (Variability Collapse):** Activations converge to their class center (within-class variance $\to 0$).
2. **NC2 (Simplex ETF):** Class centers arrange symmetrically, maximizing distance (Equiangular Tight Frame).
3. **NC3 (Self-Duality):** The linear classifier weights align perfectly with the class centers.
4. **NC4 (NCC):** Network predictions coincide with a Nearest Class Center classification.

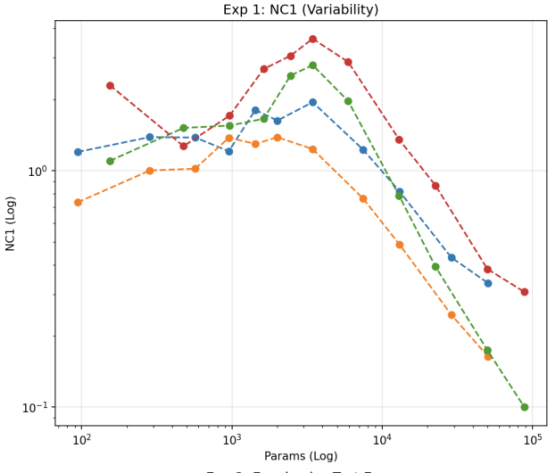
Experimental Setup

Objective: Empirically unify Double Descent, Neural Collapse, and layer similarity analysis (CKA).

- **Dataset:** MNIST-1D. Low-dimensional but morphologically complex data.
- **Injected noise:** **20% Label Noise**. Forces the network to memorize incorrect labels, exacerbating the interpolation peak.
- **Architectures:** `ResNet1D` and `StandardCNN1D`.
- **Scaling:** Variable width from $W = 2$ to $W = 64$ ($10^2 \to 10^5$ params).
- **Optimizers:** SGD vs Adam.

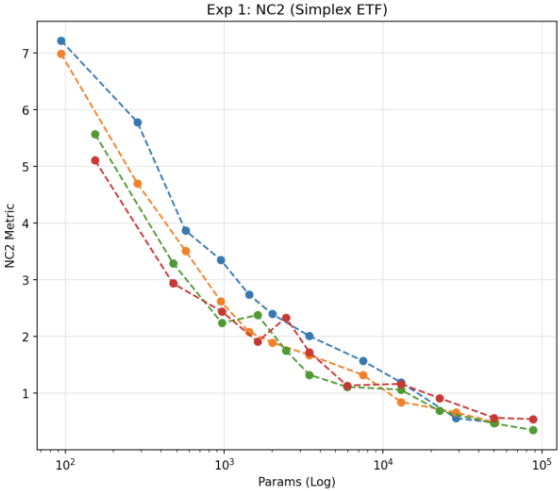
Results: The Breakdown of Collapse in the Critical Regime

- Generalization error correlates perfectly with the **NC1** metric.
- Around $10^3 - 10^4$ parameters, within-class variance explodes. The model breaks its geometry to memorize noise.
- Past the threshold ($> 10^4$), Neural Collapse is restored and the error drops.



The Impact of the Optimizer: SGD dominates Adam

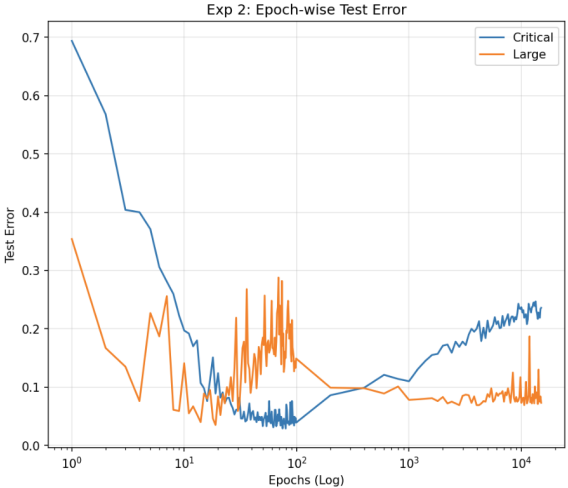
- In the "overparameterized" regime, **SGD** achieves much better geometric configurations (NC2).
- **Adam** tends to get stuck in geometrically sub-optimal local minima.
- **Why?** SGD possesses an *implicit inductive bias* that penalizes weight norms and favors wide, symmetric margins.



Training Dynamics (Epoch-wise)

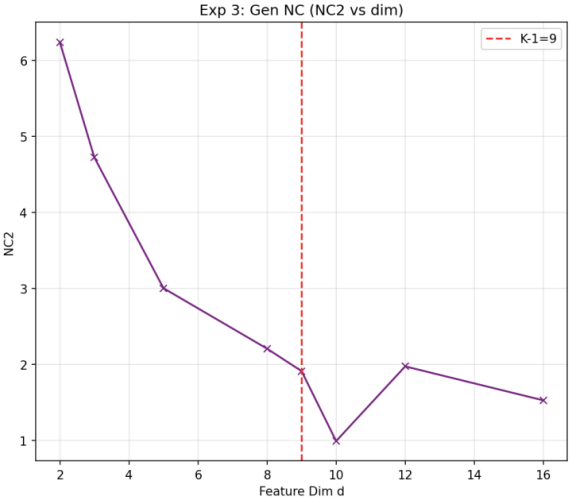
Comparison between two configurations:

- **Critical ($W = 12$):** Located on the Double Descent peak. The loss surface is highly fragmented, leading to massive error oscillations.
- **Large ($W = 64$):** Excess capacity allows rapid learning of robust features, relegating noise memorization without destroying generalization.



Dimensional Limits of Neural Collapse

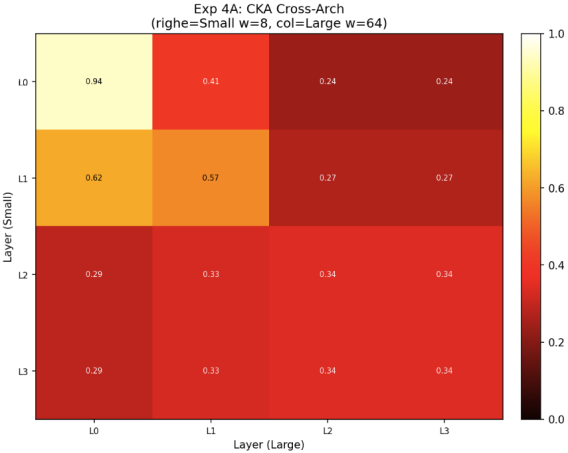
- Mathematically, a perfect Simplex ETF with K classes requires a latent dimension of $d \geq K - 1$.
- For MNIST-1D ($K = 10$), we expect collapse starting from $d = 9$.
- **Empirical Verification:** NC2 drops vertically and stabilizes at zero exactly at the theoretical limit.



CKA: Structural Differences (Small vs Large)

CKA: Measures geometric similarity $(0 \to 1)$.

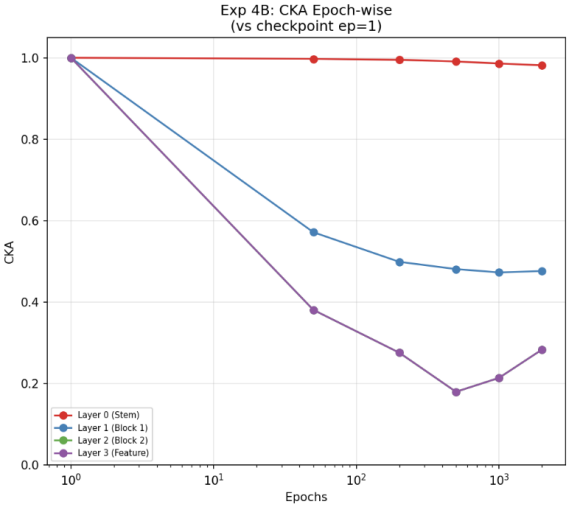
- **L0/L1:** High CKA. Critical and over-parameterized networks extract the same low-level features.
- **L3 (Deep):** CKA drops (≈ 0.34).
- The small model lacks the degrees of freedom to create the Simplex ETF, resulting in a diametrically opposed final topology.



CKA: Drift Over Time (Epoch Drift)

Comparison of activations in successive epochs relative to epoch 1:

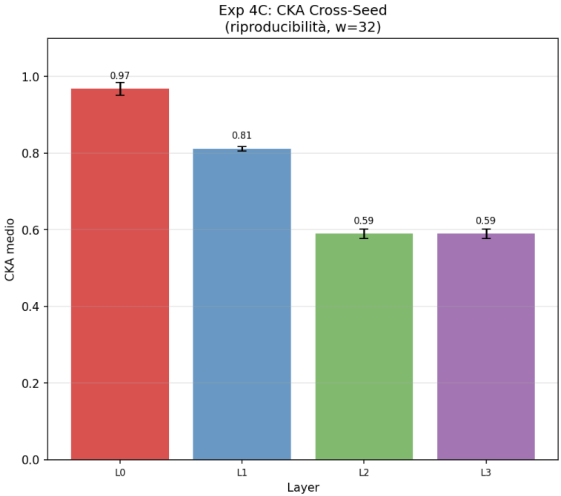
- **Stem (L0):** Converges immediately (CKA constant near 1).
- **Deep Layers (L2, L3):** Continuous drift. The latent space reorganizes over thousands of epochs, driven by the minimization of within-class variance.



CKA: Reproducibility and Geometric Degeneracy

Same model ($W = 32$), only *different initial random Seeds*:

- L0 learns deterministic features ($CKA = 0.97$).
- **L3 varies significantly ($CKA = 0.59$).**
- **Geometric Degeneracy:** There are infinite orthogonal rotations to create a perfect Simplex. Networks converge to logically isomorphic solutions, but in different absolute coordinates.



Conclusions

1. **The Empirical Link:** The Double Descent phenomenon is intimately linked to the breakdown (in the critical regime) and subsequent reconstruction (in the massive regime) of Neural Collapse.
2. **Optimization:** SGD dominates Adam, acting as an implicit regularizer that guides networks toward a perfect Simplex ETF.
3. **Deep Geometry:** CKA revealed the heterogeneity of learning. While early layers are static and deterministic, the last layer degenerates geometrically, undergoing a slow reorganization to maximize margins.

Thank you for your attention!

Questions?